Table A1
Hyperparameters for GRU4Rec. Emb. = embedding size; Hidden = hidden state size; Layers = number of GRU layers; Dropout = dropout probability; LR = learning rate.

Dataset	Epochs	Emb.	Hidden	Layers	Dropout	LR
Toys	100	64	128	1	0.5	0.01
Yelp	100	128	256	2	0.3	0.001
Books	100	128	256	2	0.3	0.0005

Table A2
Hyperparameters for SASRec. Hidden = hidden embedding size; Inner = feed-forward network dimension; Layers = number of Transformer blocks; Heads = number of attention heads; Hidden dr = hidden dropout probability; Attn dr = attention dropout probability; LR = learning rate.

Dataset	Epochs	Hidden	Inner	Layers	Heads	Hidden dr	Attn dr	LR
Toys	100	64	256	2	2	0.5	0.5	0.01
Yelp	100	128	512	2	2	0.4	0.4	0.001
Books	100	128	512	3	2	0.3	0.3	0.0005

Table A3
Hyperparameters for BERT4Rec. Hidden = hidden embedding size; Inner = feed-forward network dimension; Layers = number of Transformer blocks; Heads = number of attention heads; Hidden dr = hidden dropout probability; Attn DO = attention dropout probability; Mask = masking ratio; LR = learning rate.

Dataset	Epochs	Hidden	Inner	Layers	Heads	Hidden dr	Attn dr	Mask	LR
Toys	100	64	256	2	2	0.5	0.5	0.2	0.01
Yelp	100	128	512	2	2	0.4	0.4	0.2	0.001
Books	100	128	512	3	2	0.3	0.3	0.2	0.0005

Table A4
Hyperparameters for BPR. Emb. = embedding size; Reg. = regularization weight; LR = learning rate.

Dataset	Epochs	Emb.	Reg.	LR
Toys	100	64	10^{-4}	0.01
Yelp	100	128	10^{-4}	0.001
Books	100	128	10^{-3}	0.0005

Appendix

In this appendix, we provide the complete set of hyperparameters adopted for each model and dataset considered in our experiments. Furthermore, we extend the beyond-accuracy analysis described in Section 6.2 to the additional Books and Toys datasets, whose results were omitted from the main manuscript due to space constraints.

Hyperparameters

In the following Tables (A1-A14), we report the hyperparameter configurations used for each of the 15 evaluated methods (Pop is excluded since no parameter is needed). For each method, a separate table is provided, detailing the hyperparameters adopted across the different datasets.

Beyond-accuracy results on Books and Toys

Figures A1 and A2 report Coverage and Novelty as a function of the perturbation level for the Books and Toys datasets, respectively, extending the beyond-accuracy analysis presented in Section 6.2.

On Books, the observed trends are largely consistent with those reported for the other datasets analysed in the paper. Graph-based CF methods generally exhibit increasing coverage under sequence corruption, with NGCF showing

Table A5
Hyperparameters for DMF. User Emb. = user embedding size; Item Emb. = item embedding size; User Hidden = user MLP hidden layer sizes; Item Hidden = item MLP hidden layer sizes; LR = learning rate.

Dataset	Epochs	User Emb.	Item Emb.	User Hidden	Item Hidden	LR
Toys	100	64	64	[64,64]	[64,64]	0.01
Yelp	100	128	128	[128,64]	[128,64]	0.001
Books	100	128	128	[128,64]	[128,64]	0.0005

Table A6
Hyperparameters for SimpleX. Emb. = embedding size; Margin = contrastive loss margin; Neg. = number of negative samples; Reg. = regularization weight; Batch = training batch size; LR = learning rate; Epochs = number of training epochs.

Dataset	Epochs	Emb.	Margin	Neg.	Reg.	Batch	LR
Toys	100	64	0.8	100	10^{-4}	1024	0.01
Yelp	100	128	0.8	100	10^{-4}	4096	0.001
Books	100	128	0.8	100	10^{-3}	8192	0.0005

Table A7
Hyperparameters for LightGCN. Emb. = embedding size; Layers = number of graph convolution layers; Reg. = regularization weight; LR = learning rate.

Dataset	Epochs	Emb.	Layers	Reg.	LR
Toys	100	64	2	10^{-4}	0.01
Yelp	100	128	2	10^{-4}	0.001
Books	100	128	2	10^{-3}	0.0005

Table A8
Hyperparameters for NGCF. Emb. = embedding size; Hidden = hidden layer sizes; Node dr = node dropout probability; Message DO = message dropout probability; Reg. = regularization weight; LR = learning rate.

Dataset	Epochs	Emb.	Hidden	Node DO	Message dr	Reg.	LR
Toys	100	64	[64,64]	0.1	0.1	10^{-5}	0.01
Yelp	100	64	[64,64]	0.1	0.1	10^{-5}	0.001
Books	100	64	[64,64]	0.0	0.1	10^{-4}	0.0005

Table A9
Hyperparameters for DGCF. Emb. = embedding size; Factors = number of disentangled factors; Iter. = routing iterations; Layers = graph convolution layers; Reg. = regularization weight; Cor. = correlation regularization weight; Batch = training batch size; LR = learning rate.

Dataset	Epochs	Emb.	Factors	Iter.	Layers	Reg.	Cor.	Batch	LR
Toys	100	64	2	1	1	10^{-3}	10^{-3}	1024	0.01
Yelp	100	64	2	1	2	10^{-3}	10^{-3}	2048	0.001
Books	100	64	2	1	2	10^{-2}	5×10^{-4}	4096	0.0005

the largest increase, whereas random burst and targeted injection lead to smaller and more uniform changes. Among General CF methods, BPR shows the most pronounced coverage increase under sequence corruption, while DMF consistently achieves lower coverage and higher novelty, suggesting a tendency towards a narrower recommendation space. Sequential models maintain high and relatively stable coverage across perturbation types, although GRU4Rec shows a visible decrease under random burst. Novelty decreases under both random burst and targeted injection,

Table A10
Hyperparameters for ItemKNN. k = number of nearest neighbors; Shrink = shrinkage parameter for similarity estimation; Norm. = score normalization.

Dataset	k	Shrink	Norm.
Toys	50	0	True
Yelp	100	0	True
Books	50	10	True

Table A11
Hyperparameters for AsymKNN. k = number of nearest neighbors; α = asymmetric cosine similarity coefficient; β = score normalization coefficient; q = locality parameter; Method = neighborhood construction strategy.

Dataset	k	α	β	q	Method
Toys	50	0.3	1.0	1	User
Yelp	100	0.3	1.0	1	User
Books	50	0.3	1.0	1	User

Table A12
Hyperparameters for DeepCoNN. Word = word embedding dimension; Latent = latent representation size; Kernels = CNN kernel sizes; Reviews = maximum number of reviews per user/item; Words = maximum number of words per review; Drop = dropout rate; Batch = training batch size; LR = learning rate; Epochs = number of training epochs.

Dataset	Epochs	Word	Latent	Kernels	Reviews	Words	Batch	LR
Toys	100	300	64	[3,4,5]	10	150	1024	0.01
Yelp	100	300	64	[3,4,5]	15	200	2048	0.001
Books	100	300	64	[3,4,5]	20	150	4096	0.0005

Table A13
Hyperparameters for NARRE. Word = word embedding dimension; Latent = latent representation size; Kernels = CNN kernel sizes; Reviews = maximum number of reviews per user/item; Words = maximum number of words per review; Drop = dropout rate; Batch = training batch size; LR = learning rate; Epochs = number of training epochs.

Dataset	Epochs	Word	Latent	Kernels	Reviews	Words	Batch	LR
Toys	100	300	64	[3,4,5]	10	150	1024	0.01
Yelp	100	300	64	[3,4,5]	15	200	2048	0.001
Books	100	300	64	[3,4,5]	20	150	4096	0.0005

Table A14
Hyperparameters for TransNet. Word = word embedding dimension; Latent = latent representation size; Kernels = CNN kernel sizes; Reviews = maximum number of reviews per user/item; Words = maximum number of words per review; Drop = dropout rate; λ_{mse} = weight of the auxiliary MSE loss; Batch = training batch size; LR = learning rate; Epochs = number of training epochs.

Dataset	Epochs	Word	Latent	Kernels	Reviews	Words	λ_{mse}	Batch	LR
Toys	100	300	64	[3,4,5]	10	150	0.1	1024	0.01
Yelp	100	300	64	[3,4,5]	15	200	0.1	2048	0.001
Books	100	300	64	[3,4,5]	20	150	0.1	4096	0.0005

reflecting a popularity-drift effect similar to that observed for other ranking-based models. Classical baselines remain largely unchanged across noise types and levels, for both coverage and novelty, due to their reliance on static recommendation strategies.

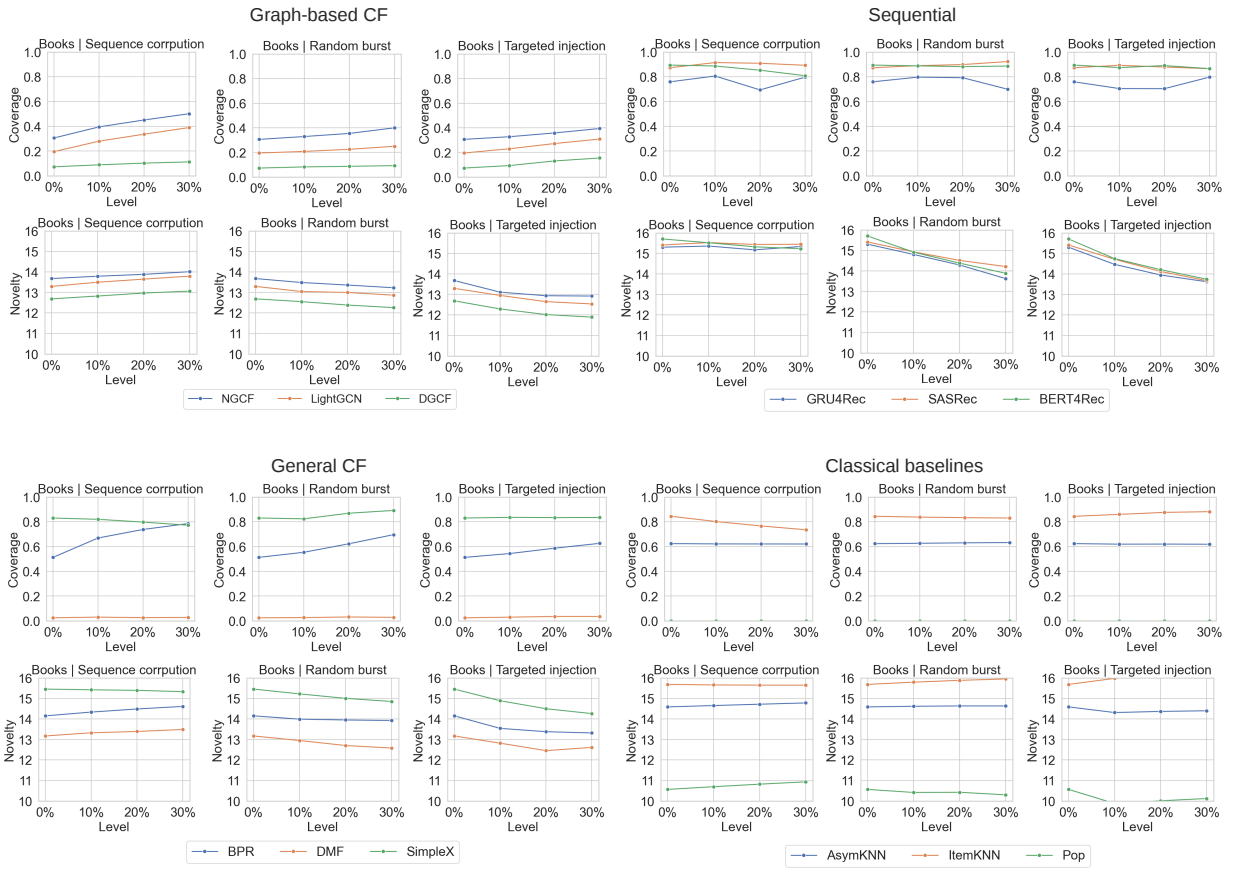


Figure A1: Coverage and Novelty as a function of the perturbation level (0%–30%) on Books, grouped by model family (Graph-based CF, Sequential, General CF, Classical baselines), under three perturbation types: sequence corruption, random burst, and targeted injection.

On Toys, the same general model-family behaviours can be observed, although the effects are noisier and less monotonic. In particular, BPR exhibits a decrease in coverage under sequence corruption, opposite to the trend observed on Books, highlighting that the impact of perturbations depends on dataset-specific characteristics such as catalogue size and popularity distribution. Graph-based CF models also display less stable trends across noise levels compared with Books, suggesting a weaker structural signal under the injected perturbations. Sequential models maintain higher coverage than other approaches, with SASRec outperforming GRU4Rec and BERT4Rec, while novelty again decreases under random burst and targeted injection. Classical baselines (Pop, ItemKNN, and AsymKNN) remain largely unaffected by perturbations, confirming their robustness to the considered noise injections.

Overall, these results confirm that the qualitative differences among model families are preserved across datasets: classical baselines remain largely invariant, graph-based methods are sensitive to structural perturbations, and sequential models are affected by synthetic-user injections. However, the magnitude and direction of coverage and novelty variations remain dataset-dependent, with Toys exhibiting less stable patterns than Books.

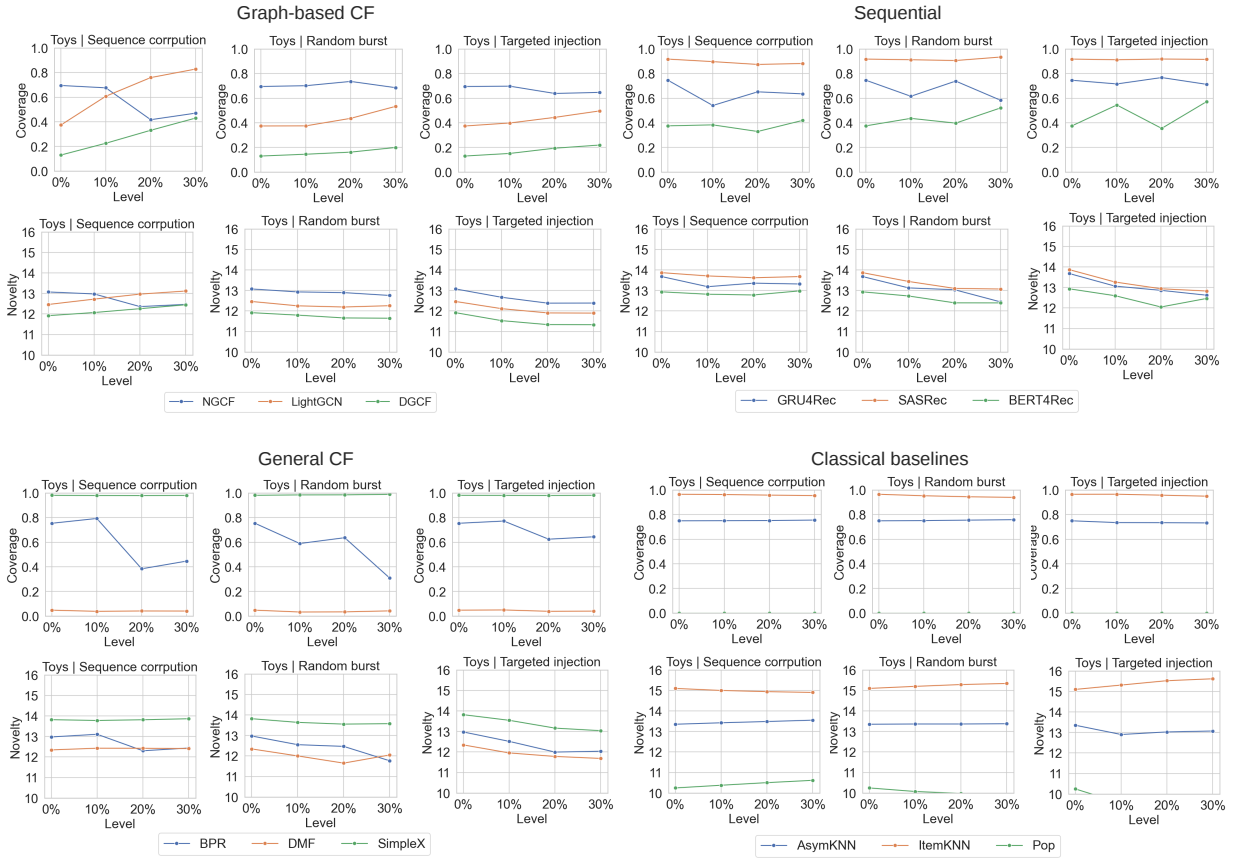


Figure A2: Coverage and Novelty as a function of the perturbation level (0%–30%) on Toys, grouped by model family (Graph-based CF, Sequential, General CF, Classical baselines), under three perturbation types: sequence corruption, random burst, and targeted injection.